

NEWS INTELLIGENCE MONITOR - v4.1



Building a Production-Ready MLOps Platform for News Classification

From experimentation to industrialization — an end-to-end NLP platform covering the complete MLOps lifecycle

❏ This is not a Machine Learning notebook. It is an industrial AI platform.

Why Article Classification at Scale

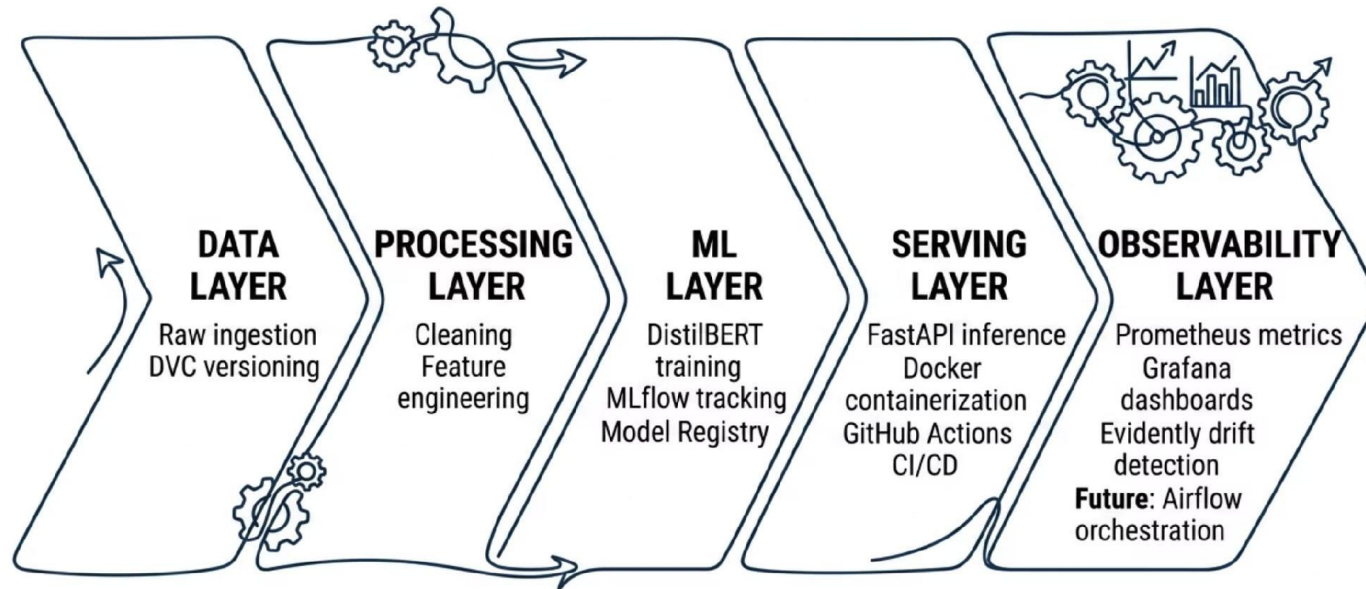
The operational problem behind the engineering



Manual categorization doesn't scale. Automated classification unlocks structured knowledge pipelines, editorial tooling, and downstream semantic search — the foundation for a production RAG system.

Platform Architecture

Layered enterprise design — separation of concerns at every tier



Engineering Decisions

Every choice is deliberate — no accidental complexity



DistilBERT

97% of BERT performance at 40% fewer parameters — optimal inference latency for production serving



Docker

Environment parity from dev to prod — eliminates model drift caused by dependency mismatch



Prometheus + Grafana

Time-series metrics collection and visualization — real-time model health observability



FastAPI

Async-native, OpenAPI auto-documentation, sub-millisecond overhead — production API tier



MLflow

Experiment lineage, artifact versioning, model registry — full reproducibility contract

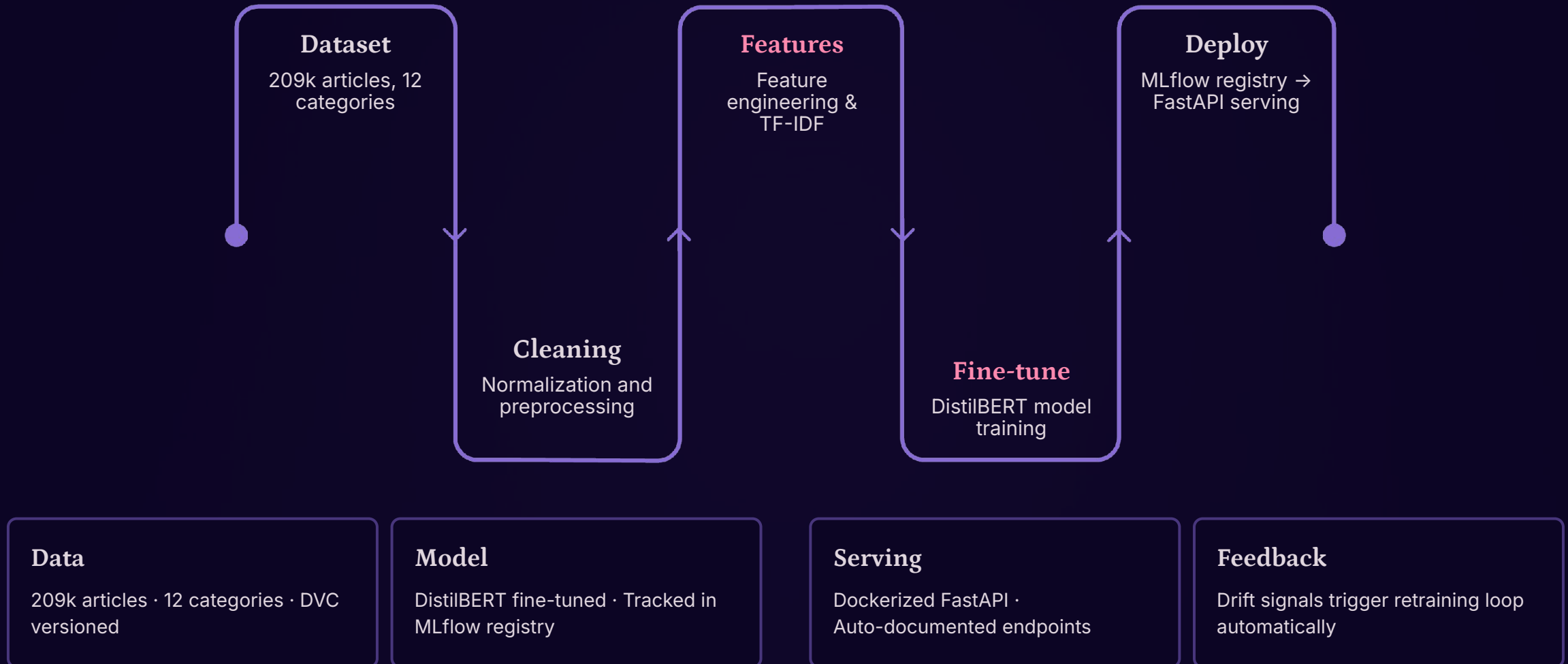


Evidently + DVC

Statistical drift detection + data versioning — the closed loop for continuous improvement

ML Lifecycle Pipeline

End-to-end, traceable, reproducible



Observability Stack

You cannot improve what you cannot measure

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Prometheus
Scrapes inference metrics, latency, request counts

Grafana
Real-time dashboards, alerting rules

Evidently
Data drift reports, feature distribution shifts, PSI scores

Retraining Decision
Automated trigger when drift threshold exceeded

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The Quick Brown Fox Jumps Over The Lazy Dog.

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Prometheus

Scrapes prediction confidence, latency p50/p95, request volume

Grafana

Live dashboards — model health, throughput, error rates

Evidently

Feature distribution shift · PSI scores · Automated drift reports

Retraining Loop

Drift threshold breach → triggered pipeline → new model candidate

Platform Deliverables

What was actually shipped

209K

Articles

Versioned dataset · 12 categories · DVC tracked

12

Categories

Multi-class classification · balanced evaluation

5

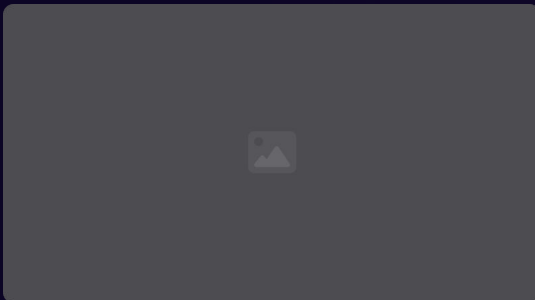
Services

FastAPI · MLflow · Prometheus · Grafana · Evidently

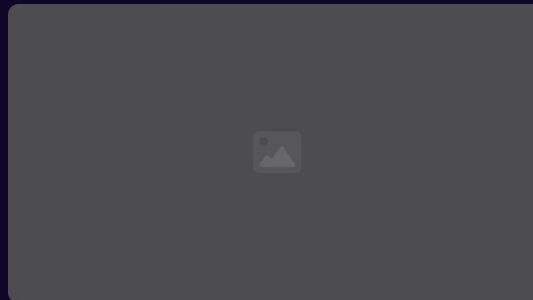
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Docker Stack

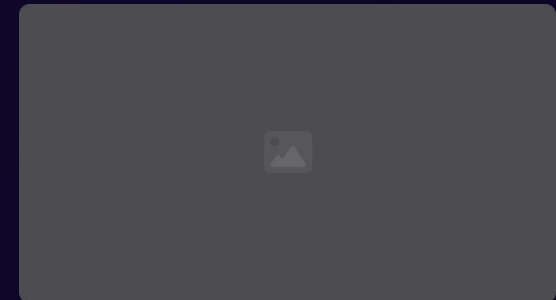
Single compose deployment · full environment parity



FastAPI — Swagger UI · auto-documented
inference endpoints



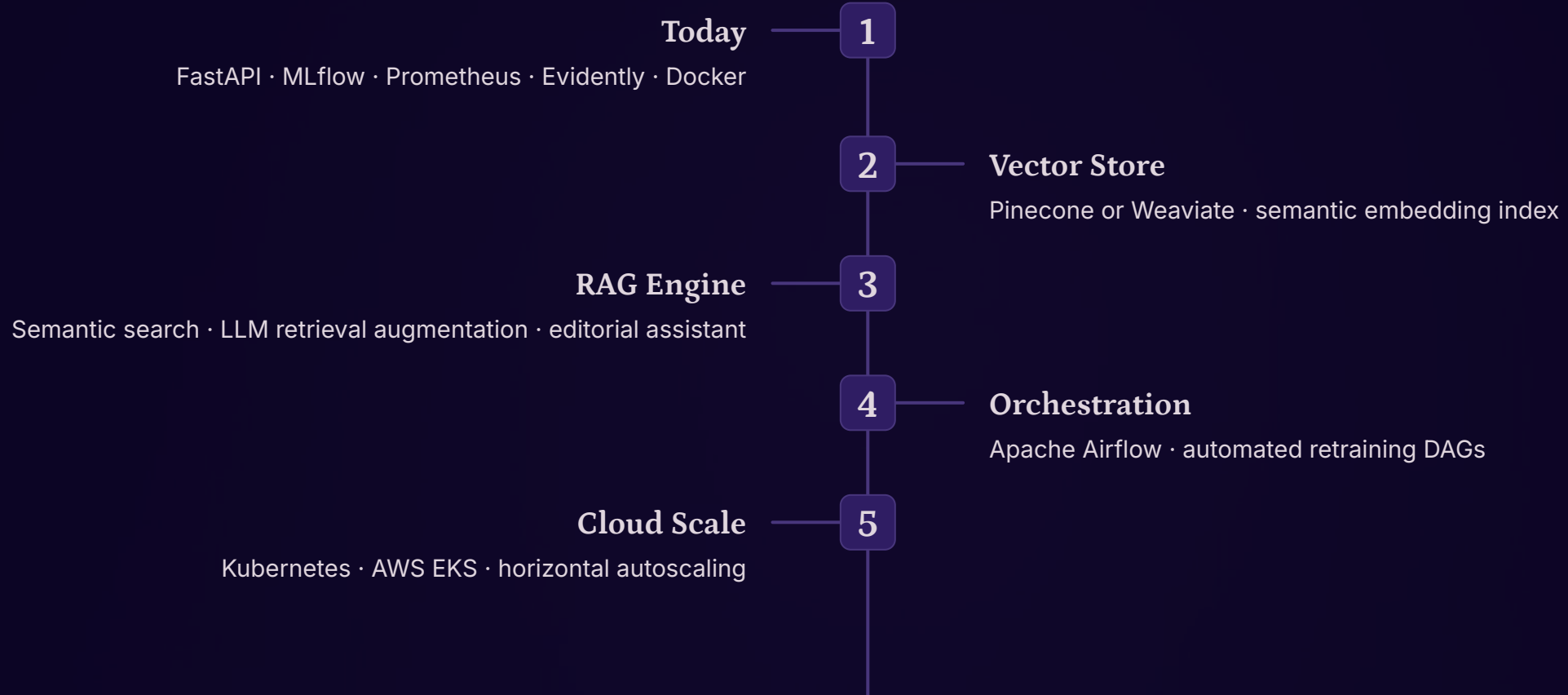
MLflow — experiment runs, model registry, artifact
lineage



Grafana — live observability dashboard, drift alerts

Strategic Roadmap

Architecture designed for evolution — not refactoring



Every architectural layer was chosen to be replaced or extended — not locked in.



Beyond Machine Learning

Reproducibility

DVC + MLflow · every experiment is traceable

Observability

Metrics, drift, health — visible at all times

Scalability

Stateless services · container-native · cloud-ready

Maintainability

CI/CD · versioned data · modular architecture

Building a model is only the beginning. Building a **reliable AI platform** is the real engineering challenge.